中国科学技术大学计算机学院

《数字电路实验》报告

实验题目：实验 07 FPGA 实验平台及 IP 核使用

学生姓名：王琛

学生学号：PB20111696

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计算机实验教学中心制

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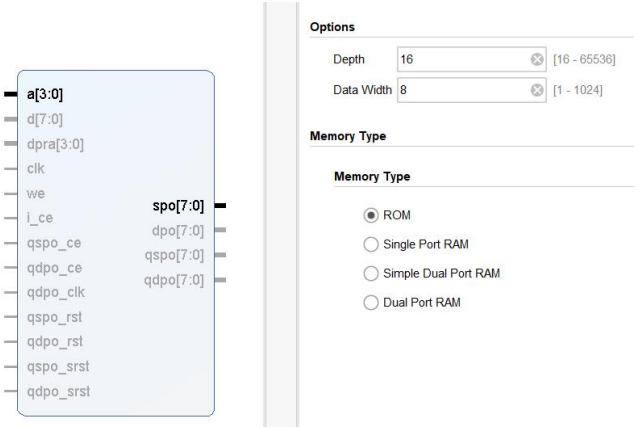
【实验题目】

07 FPGA 实验平台及 IP 核使用

【实验练习】

题目 1

首先例化 16*8bit 的 ROM



初始化如下

COE File Editor - practice1.coe

Key	Value
memory_initialization_radix	16
memory_initialization_vector	3F 06 5B 4F 66 6D 7D 07 7F 6F 77 7C 39 5E 79 71

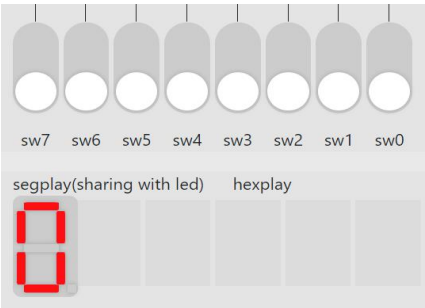
在设计文件中，将开关和 led 分别接入 ROM 的输入输出端口

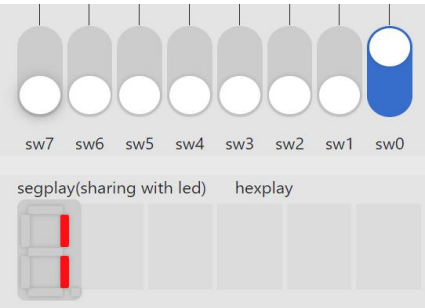
```
module practice1(
input [3:0] sw,
output [7:0] led);
dist_mem_gen_0 dist_mem_gen_0(sw,led);
endmodule
```

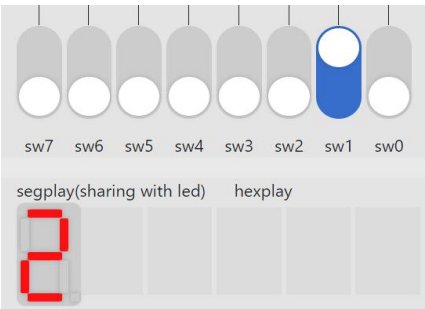
在约束文件中将开关和 led 接入对应管脚

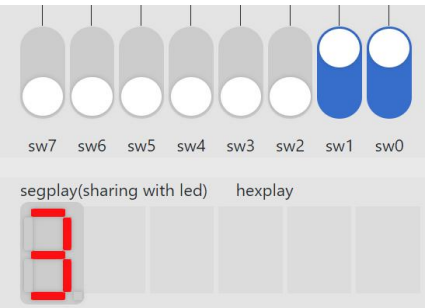
```
set_property -dict { PACKAGE_PIN D14 IOSTANDARD LVCMOS33 } [get_ports { sw[0] }];
set_property -dict { PACKAGE_PIN F16 IOSTANDARD LVCMOS33 } [get_ports { sw[1] }];
set_property -dict { PACKAGE_PIN G16 IOSTANDARD LVCMOS33 } [get_ports { sw[2] }];
set_property -dict { PACKAGE_PIN H14 IOSTANDARD LVCMOS33 } [get_ports { sw[3] }];
## Leds
set_property -dict { PACKAGE_PIN G18 IOSTANDARD LVCMOS33 } [get_ports { led[7] }];
set_property -dict { PACKAGE_PIN F18 IOSTANDARD LVCMOS33 } [get_ports { led[6] }];
set_property -dict { PACKAGE_PIN E17 IOSTANDARD LVCMOS33 } [get_ports { led[5] }];
set_property -dict { PACKAGE_PIN D17 IOSTANDARD LVCMOS33 } [get_ports { led[4] }];
set_property -dict { PACKAGE_PIN G17 IOSTANDARD LVCMOS33 } [get_ports { led[3] }];
set_property -dict { PACKAGE_PIN E18 IOSTANDARD LVCMOS33 } [get_ports { led[2] }];
set_property -dict { PACKAGE_PIN D18 IOSTANDARD LVCMOS33 } [get_ports { led[1] }];
set_property -dict { PACKAGE_PIN C17 IOSTANDARD LVCMOS33 } [get_ports { led[0] }];
```

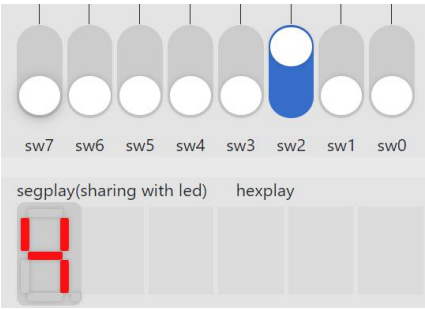
在 FPGAOL 平台上测试如下

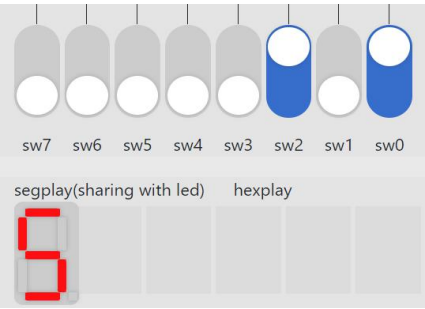


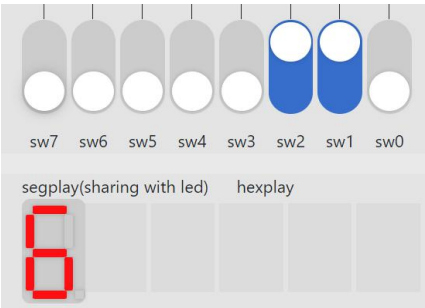


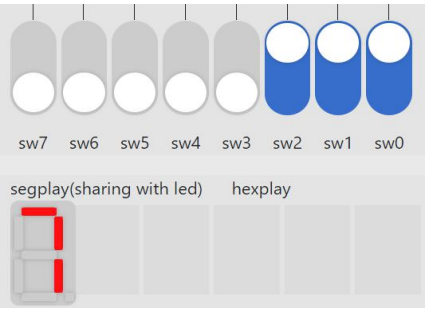


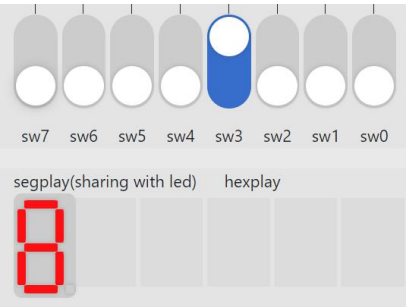


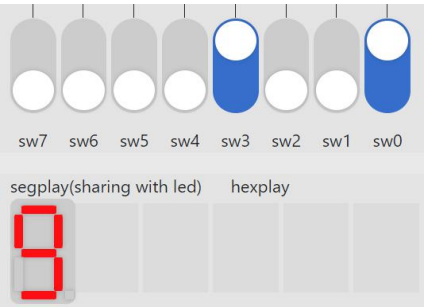












题目 2

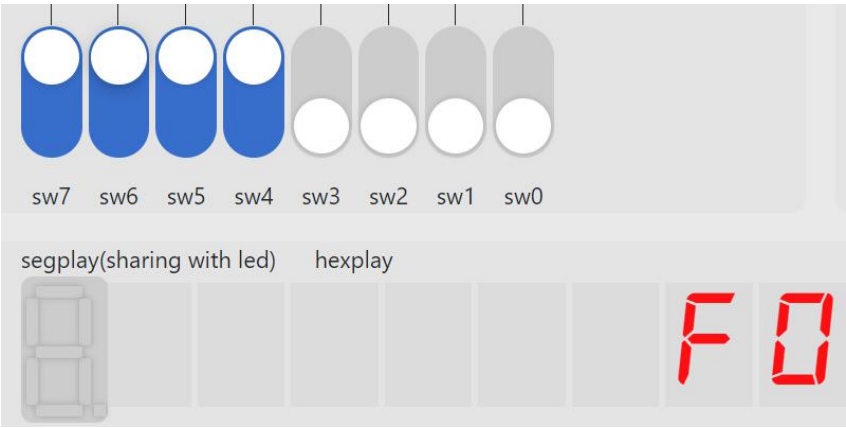
整个扫描周期为 50HZ，有两个需要显示的数码管，需要 100HZ 的时钟信号，故使用计数器产生低频脉冲信号，设计文件如下

```
module practice2(
    input clk,
    input [7:0] sw,
    output reg [2:0] hexplay_an,
    output reg [3:0] hexplay_data
);

reg [32:0] hexplay_cnt;

always@(posedge clk)
    if (hexplay_cnt >= (2000000 / 2)) hexplay_cnt <= 0;
    else hexplay_cnt <= hexplay_cnt + 1;

always@(posedge clk)
    if (hexplay_cnt == 0)
        if(hexplay_an >= 1) hexplay_an <= 0;
        else hexplay_an <= 1;
    else;

always@(*) begin
    case(hexplay_an)
        0: hexplay_data = sw[3:0];
        1: hexplay_data = sw[7:4];
    endcase
end
```

约束文件如下

```
# Clock signal
set_property -dict { PACKAGE_PIN E3    IOSTANDARD LVCMOS33 } [get_ports { clk }];

# FPGAOL SWITCH

set_property -dict { PACKAGE_PIN D14   IOSTANDARD LVCMOS33 } [get_ports { sw[0] }];
set_property -dict { PACKAGE_PIN F16   IOSTANDARD LVCMOS33 } [get_ports { sw[1] }];
set_property -dict { PACKAGE_PIN G16   IOSTANDARD LVCMOS33 } [get_ports { sw[2] }];
set_property -dict { PACKAGE_PIN H14   IOSTANDARD LVCMOS33 } [get_ports { sw[3] }];
set_property -dict { PACKAGE_PIN E16   IOSTANDARD LVCMOS33 } [get_ports { sw[4] }];
set_property -dict { PACKAGE_PIN F13   IOSTANDARD LVCMOS33 } [get_ports { sw[5] }];
set_property -dict { PACKAGE_PIN G13   IOSTANDARD LVCMOS33 } [get_ports { sw[6] }];
set_property -dict { PACKAGE_PIN H16   IOSTANDARD LVCMOS33 } [get_ports { sw[7] }];

# FPGAOL HEXPLAY

set_property -dict { PACKAGE_PIN A14   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[0] }];
set_property -dict { PACKAGE_PIN A13   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[1] }];
set_property -dict { PACKAGE_PIN A16   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[2] }];
set_property -dict { PACKAGE_PIN A15   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[3] }];
set_property -dict { PACKAGE_PIN B17   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[0] }];
set_property -dict { PACKAGE_PIN B16   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[1] }];
set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[2] }];
```

在 FPGAOL 平台进行简单测试



题目 3

使用计数器产生 10HZ 的低频脉冲来更新计时器，产生 200HZ 的低频脉冲来将计时器的数值显示在数码管上，设计文件如下

```
module practice3(
    input clk,
    input rst,
    output reg [2:0] hexplay_an,
    output reg [3:0] hexplay_data
);

reg [15:0] data;
reg [32:0] hexplay_cnt;

initial
    data=0;

always@(posedge clk) begin
    if (hexplay_cnt >= (2000000 / 4)) hexplay_cnt <= 0;
    else hexplay_cnt <= hexplay_cnt + 1;
end
```

```
always@(posedge clk) begin
    if (hexplay_cnt == 0)
        if (hexplay_an == 3) hexplay_an <= 0;
        else hexplay_an <= hexplay_an + 1;
    else ;
end

always@(*) begin
    case(hexplay_an)
        0: hexplay_data = (data[3:0] <= 4'd9) ? data[3:0] : 0 ;
        1: hexplay_data = (data[7:4] <= 4'd9) ? data[7:4] : 0;
        2: hexplay_data = (data[11:8] <= 4'd5) ? data[11:8] : 0;
        3: hexplay_data = data[15:12];
    endcase
end

reg [31:0] timer_cnt;
always@(posedge clk) begin
    if (timer_cnt >= 10000000) timer_cnt <= 0;
    else timer_cnt <= timer_cnt + 1;
end
```

```
always@(posedge clk) begin
    if (timer_cnt == 0) begin
        if (rst) begin
            data[15:12] <= 4'd1;
            data[11:8] <= 4'd2;
            data[7:4] <= 4'd3;
            data[3:0] <= 4'd4;
        end
        else begin
            data[3:0] <= data[3:0] + 1;
            if(data[3:0] >= 10) begin
                data[3:0] <= 0;
                data[7:4] <= data[7:4] + 1;
            end
            else ;
            if(data[7:4] >= 10) begin
                data[7:4] <= 0;
                data[11:8] <= data[11:8] + 1;
            end
            else ;
            if(data[11:8] >= 6) begin
                data[11:8] <= 0;
                data[15:12] <= data[15:12] + 1;
            end
            else ;
        end
    end
end

endmodule
```

约束文件如下

```
# Clock signal
set_property -dict { PACKAGE_PIN E3    IOSTANDARD LVCMOS33 } [get_ports { clk }];

# FPGAOL HEXPLAY

set_property -dict { PACKAGE_PIN A14   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[0] }];
set_property -dict { PACKAGE_PIN A13   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[1] }];
set_property -dict { PACKAGE_PIN A16   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[2] }];
set_property -dict { PACKAGE_PIN A15   IOSTANDARD LVCMOS33 } [get_ports { hexplay_data[3] }];
set_property -dict { PACKAGE_PIN B17   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[0] }];
set_property -dict { PACKAGE_PIN B16   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[1] }];
set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports { hexplay_an[2] }];

# FPGAOL BUTTON & SOFT_CLOCK

set_property -dict { PACKAGE_PIN B18   IOSTANDARD LVCMOS33 } [get_ports { rst }];
```

在 FPGAOL 平台测试如下

【总结与思考】

1. 请总结本次实验的收获

(1)学会了 FPGAOL 在线实验平台结构及使用

(2)学习了 FPGA 开发各关键环节

(3)了解了 IP 核（知识产权核）

2. 请评价本次实验的难易程度

本次实验题目较简单

3. 请评价本次实验的任务量

实验练习能够在规定时间内完成

4. 请为本次实验提供改进建议

无